

Sheet 8

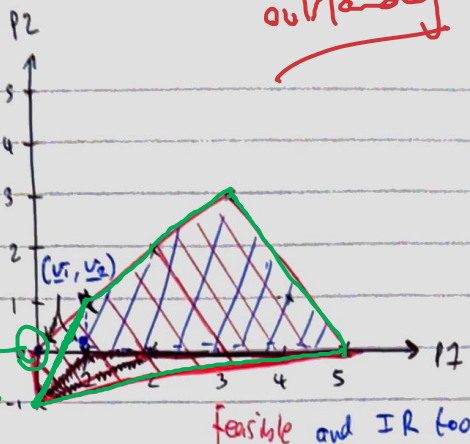
outstanding

1.

	l	m	r
T	4,1	2,2	1,0
M	5,0	3,3	1,1
B	1,0	2,0	0,-1

a)

(0,0) is not feasible.



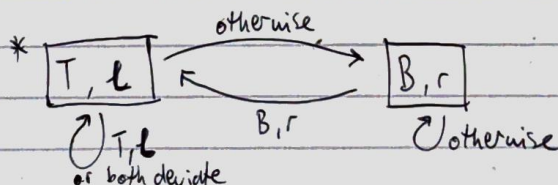
- b) P1 can hold P2 to $\underline{v}_2 \leq 0$ by playing B; P2 can guarantee $\underline{v}_2 \geq 0$ by l or m. So $\underline{v}_2 = 0$.

Similarly $\underline{v}_1 \leq 1$ with r; $\underline{v}_1 \geq 1$ with T or M, so $\underline{v}_1 = 1$.

- c) ~~IESDS~~ IESDS: eliminate r < m. Now T < M, B < M. So PSNE is (M,m). And since IESDS doesn't remove any NE and only one strategy remains for each, we're done, no MSNE.

- d) One-shot NE has payoffs (3,3), so (4,2) doesn't Pareto dominate and indeed ^{utilization} social welfare is lower, but P2 better-off.

(4,1) is in the set of feasible and IR payoffs, so we can sustain it in SPE by Perfect Folk Theorem. But actually, consider a simpler stick-carrot scheme:



To avoid ~~unilateral~~ deviations We need to check one-shot deviation principle at each state.

$\boxed{T, l}$ P2: $5 + 0\delta \leq 4 + 4\delta \Rightarrow \delta \geq 1/4$ as best deviation would be to M
P2: dev. to m, so need $2 + -1\delta \leq 1 + 1\delta \Rightarrow \delta \geq 1/2$

$\boxed{B, r}$ P2: dev. to T or M. $1 + 0\delta \leq 0 + 4\delta \Rightarrow \delta \geq 1/4$
P2: dev. to l or m. $0 + -\delta \leq -1 + \delta \Rightarrow \delta \geq 1/2$

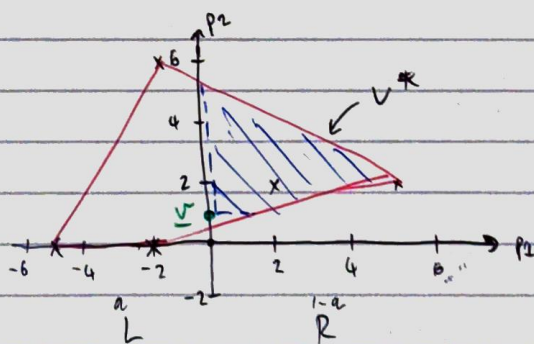
So we can sustain T, l as outcome for $\delta \geq 1/2$.

- e) It's true that repeated games do make coordination / cooperation easier - the threat of punishment into the future means players can sustain more outcomes as NE (and SPE!) than otherwise. So a given cooperative outcome may be possible when not in stage game.

Great
But it doesn't really make sense to say that cooperation is more likely. Folk Theorems don't tell you which eq^{*} will be sustained; indeed they usually have the result that a vast number are possible. And in this question, (T,L) is strictly worse for p2 than stage game PSNE, so it's not really "cooperation" in the usual sense at all; more like coercion.

- 2.a) If p2 mixes L and R w.p. $\frac{1}{2}$ she can hold $\underline{v}_1 \leq 0$. And p1 can guarantee $\underline{v}_1 \geq 0$ with $\frac{2}{7}T + \frac{5}{7}M$. So $\underline{v}_1 = 0$.

For p2's minmax: p2 can hold her to $\underline{v}_2 \leq 1$ with $\frac{1}{2}T + \frac{1}{2}M$. And if p2 mixes $\frac{1}{2}L + \frac{1}{2}R$ she guarantees $\underline{v}_2 \geq 1$. So $\underline{v}_2 = 1$.



b)

p	T	<u>5, 2</u>	-5, 0.
1-p	M	-2, 0	<u>2, 2</u>
B	-1, 6	-1, 6	

B is FESD by $\frac{2}{7}T + \frac{5}{7}M$.

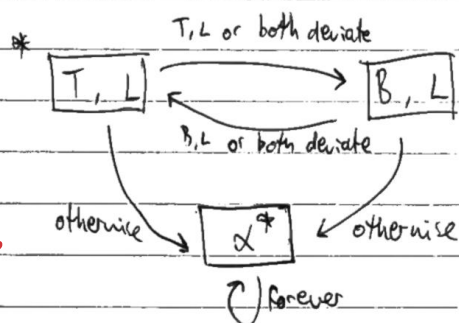
Two PSNE: (T,L) and (M,R). Also one MSNE (both must mix):

$$\frac{1}{2} 2p = 2(1-p) \Rightarrow p = \frac{1}{2}$$

$$5q - 5(1-q) = -2q + 2(1-q) \Rightarrow q = \frac{7}{14} = \frac{1}{2}$$

$$\text{so } (\frac{1}{2}T + \frac{1}{2}M, \frac{1}{2}L + \frac{1}{2}R) =: \alpha^*$$

c) we want to alternate between (T, L) and (B, L) , on-path



Here a grim trigger with Nash reversion to MSNE should work, since it's strictly worse than the average payoff from on-path for both.

Nice

We know this is sustainable as SPE by Friedman's theorem directly, because x^* forever is clearly NE in all subgames given it's NE in stage game, and given P-dominated by target outcome, will be worse in PV for $\delta \rightarrow 1$. If payoffs $(0, 1)$.

But to check with one-shot deviation principle:

(x^*) : NE, so no deviations unilaterally for sure, regardless of δ .

(T, L) : likewise.

(B, L) : P2 doesn't have strict incentive to deviate. For P2 to comply we need her to not deviate to T, so

$$5 + 0 \leq -1 + 5\delta - \delta^2 + 5\delta^3 - \dots$$

$\uparrow$
MSNE avg

$$= \frac{-1 + 5\delta}{1 - \delta^2}$$

$$\Rightarrow 6 - 5\delta^2 - 5\delta^3 \leq 0, \quad \left(\delta^* + \frac{1}{2}\right)^2 - \frac{6}{5} \delta^* - \frac{1}{4} \geq 0$$

Great

so $\delta \geq 0.70$ works.

d) SPE by backward induction. In final period both must play Nash. For $v_2 = 8$, we need to get 6 then 2 given B not part of any NE, so can't be played in final period.
 can get 50 at $t=1$ p1 will get -1, then at $t=2$ need E payoff of 2.5, achieved by 50:50 mix between the PSNE.
 Achieve that with a correlating device (public coin toss).

So we have

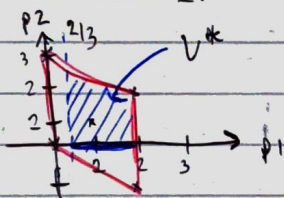
if (B,L) same outcome but temptation for P2 too great, as deviates to T.

- t=1: Play (B,R) ←
- t=2: On-path: flip a coin. If heads, play (T,L) else (M,R).
 Off-path: we need some punishment to ensure (B,R) played at t=1.
- P2 doesn't have strict incentive to deviate anyway, so if she does let's just say we play (T,L), but doesn't matter.
 - P2 might deviate to M, for payoff 2. If the punishment is the MSNE then her t=2 payoff is 0, so overall worse to deviate as required (2 vs 2.5).
- If both deviate, doesn't really matter, but say (T,L).

Independently randomize at t=1 but use the realized outcome at t=1 to correlate actions at t=2.

History is your correlation device

	L	R
U	1, 1	0, 0
M	0, 3	2, 2
D	0, 0	2, -1



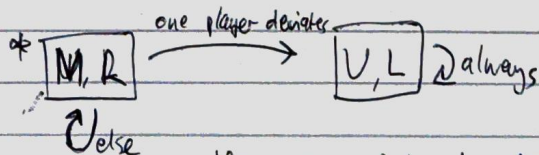
a) $\underline{v}_1 \geq 2/3$ by $\frac{2}{3}U + \frac{1}{3}M$ and $\leq 2/3$ by $\frac{2}{3}L + \frac{1}{3}R$
 so $\underline{v}_1 = 2/3$ ✓

$\underline{v}_2 \geq 0$ by L and ≤ 0 by D.
 so $\underline{v}_2 = 0$ ✓

b) L strictly dominates R. So (U,L) unique NE; (1,1) payoffs

c) we can just use Friedman's theorem with Nash reversion (grim trigger).

Target payoffs (2,2) starts to dominate the PSNE (U,L).



Use one-shot deviation principle.

MR

P2 wouldn't deviate provided at all since no strict incentive.

P2 doesn't (to L) provided $3 + \delta + \delta^2 + \dots \leq 2/(1-\delta) \Rightarrow \delta \geq 1/2$ ✓

U,L

is NE so no worries here, for either player.

$3 + \frac{\delta}{1-\delta} \leq \frac{2}{1-\delta} \Rightarrow 3 - 2\delta \leq 2 \Rightarrow 2\delta \geq 1$

d) Previous grim trigger wouldn't work, not sufficiently patient to deter p2.

Suppose instead p^1 would minimize p^2 for some number of periods, s then return to $(14, 2)$. If $s=2$ we need

$$3 + 0 + \delta^2 + \dots \leq 2/(1-\delta)$$

$$\Rightarrow 3 + \frac{\delta^2}{1-\delta} \leq 2/(1-\delta)$$

$$3 - 3\delta + \delta^2 \leq 2 \quad \text{which has LHS} = 2\frac{1}{9} \text{ for } \delta = \frac{1}{3}.$$

So we have to let $\gamma = 2$, get

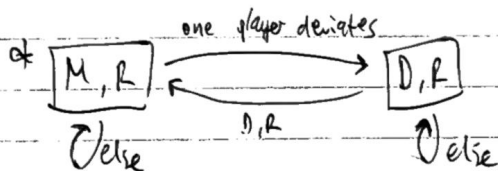
And actually this just won't work for any s because we'll already have $3-3J=2$ binning and the J^{s+2} term strictly +ve.

So we need an even more punishment which will have to be $(0, R)$.

Then $3 - \delta + \overbrace{\delta^2 + \dots}^{\delta^2 / 1 - \delta} \leq 2 / 1 - \delta$ required

$$3 - 3\delta - \delta + \delta^2 + \delta^2 \leq 2$$

$$= 3 - 4\delta + 2\delta^2 \leq 2 \quad \text{where LHS} = 1^q/q \text{ and this works}$$



One-Shot derivation principle:

M, R - Already shown p2 won't deviate.

- As before, ^{strict} no incentive for $p \geq 1$ to.

$[D, R]$ - $P2$ has no strict incentive to deviate.

- p_2 could deviate to L for 0, then -1 ^{as planned} So we need

OK $-\delta \leq -1 + 2\delta \Rightarrow \delta \geq 1/3$ satisfied.

So it's SDE.

Excellent